

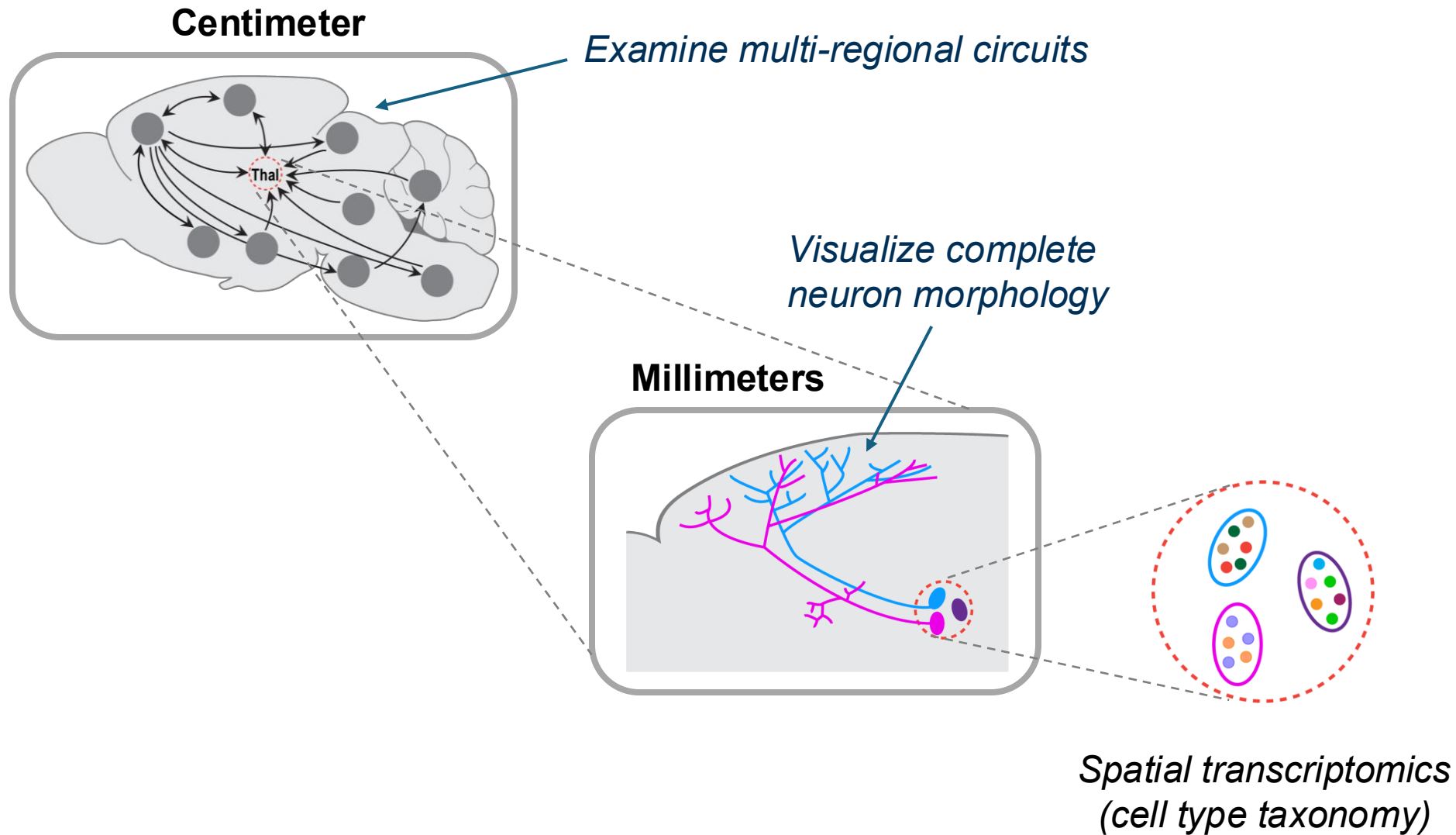


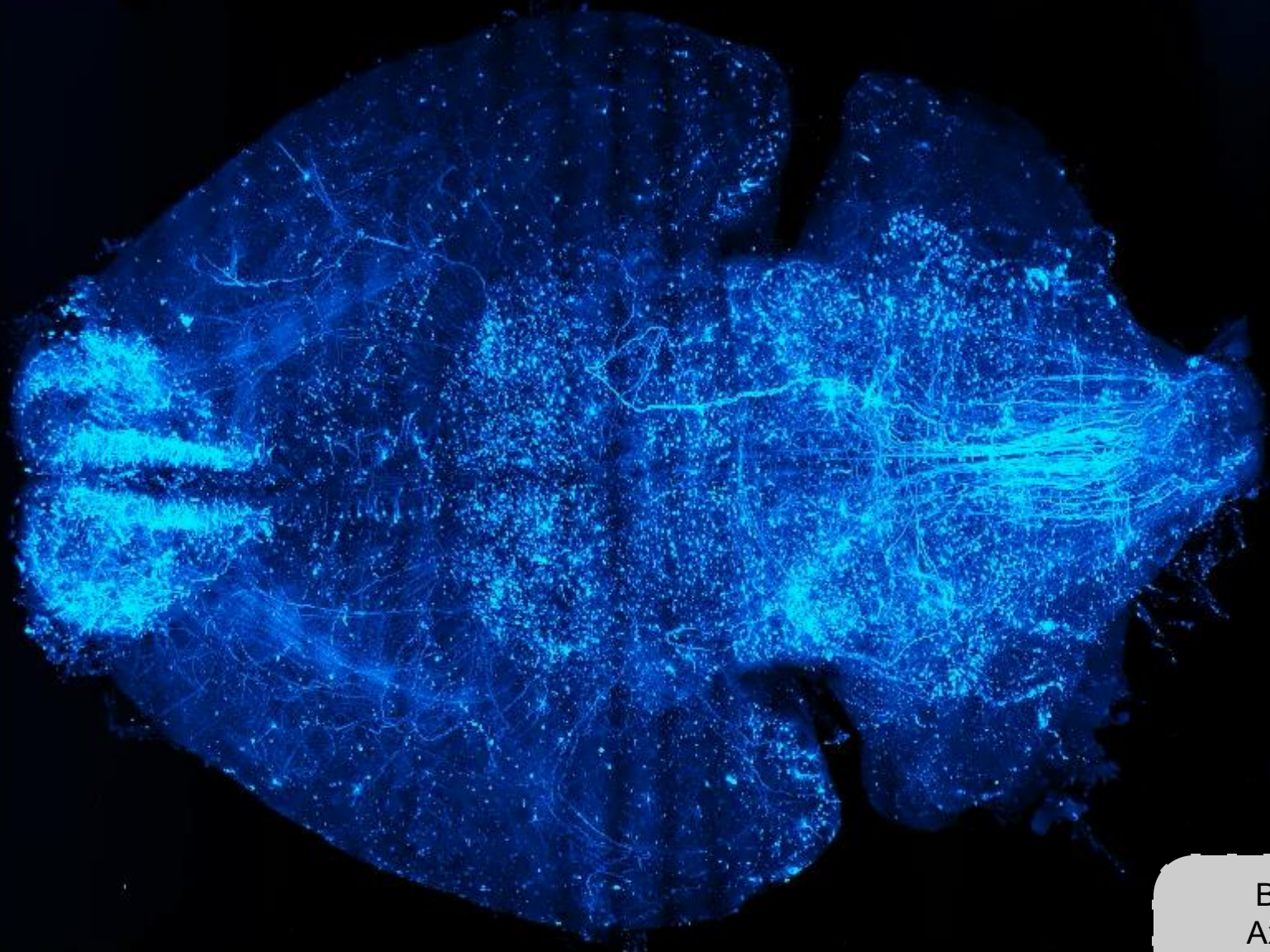
ALLEN INSTITUTE *for*
NEURAL DYNAMICS

CHARACTERIZATION OF A NEW EXPANSION-ASSISTED SELECTIVE PLANE ILLUMINATION MICROSCOPE (EXA-SPIM)

KAELIN WULF

Mapping brain-wide neural circuits



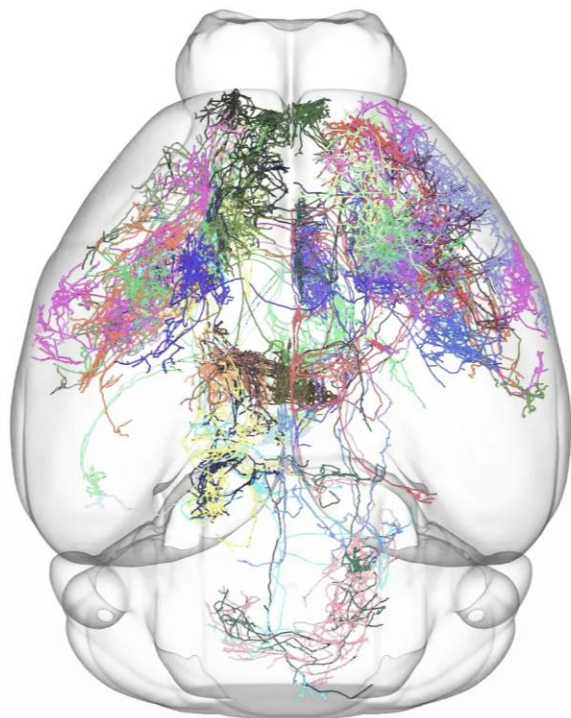


5 mm

Brain volume = 0.5 cm^3
Axon diameter = 100 nm
Target resolution = 250 nm
Data size = $64 \text{ TB} \rightarrow 100 \text{ TB (!)}$

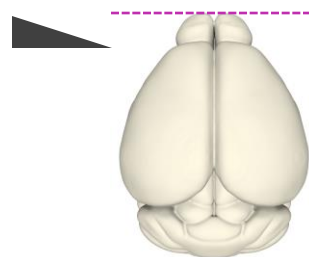
Challenge - brain-wide reconstruction of individual neurons

Mapping circuits
(MouseLight, fMOST)



centimeter scale with
nanoscale resolution

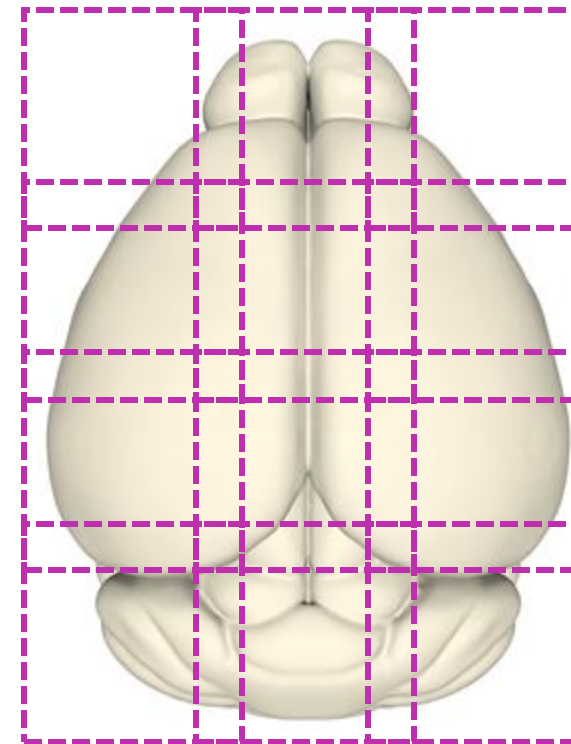
Prior approaches:
sectioning-based



long imaging times
poor axial resolution
many tiles

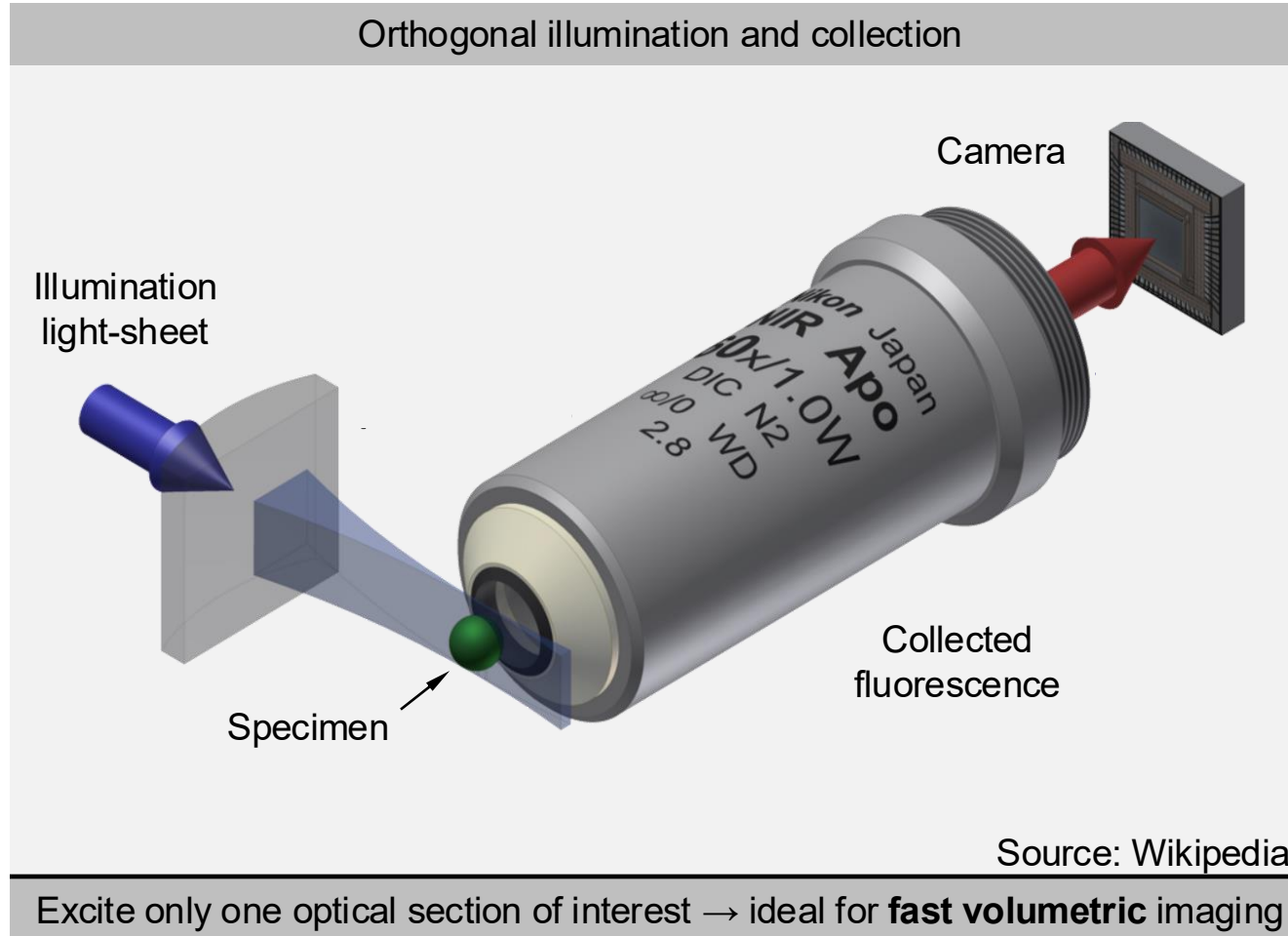
EXPAND

New approach:
light-sheet microscopy + expansion



short imaging times
better axial resolution
few tiles

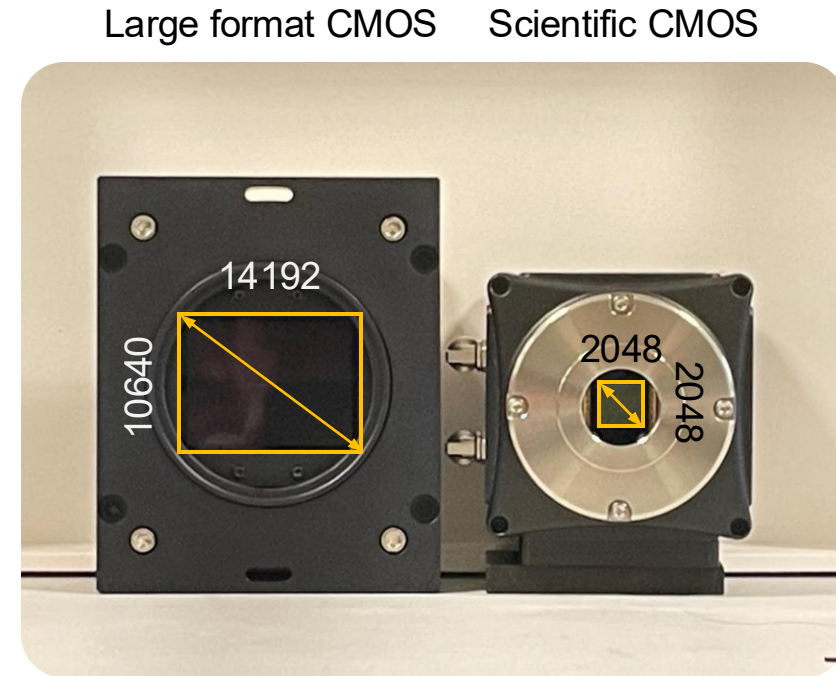
What is light-sheet microscopy?



Microscopy wish list...

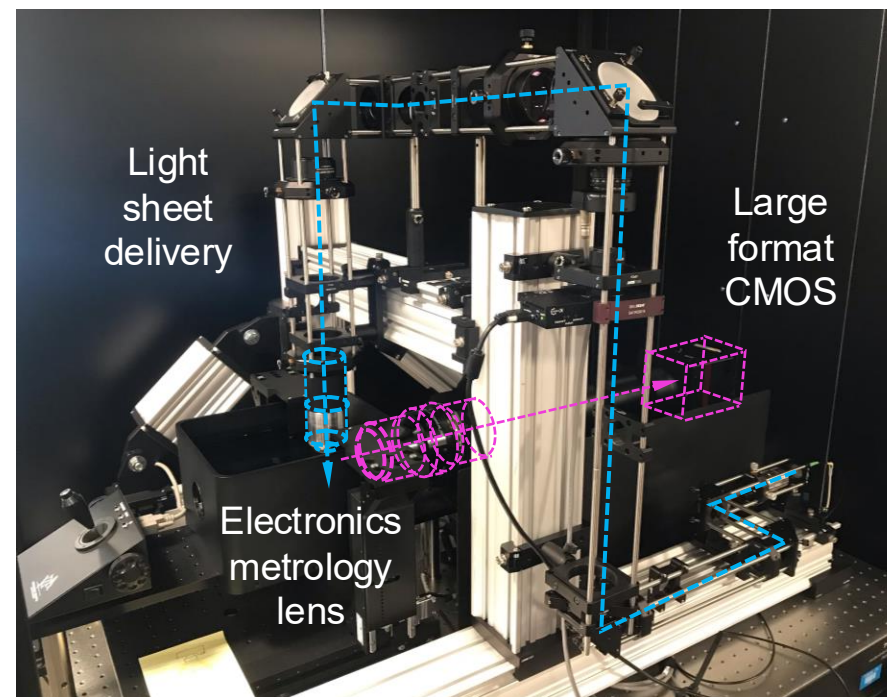
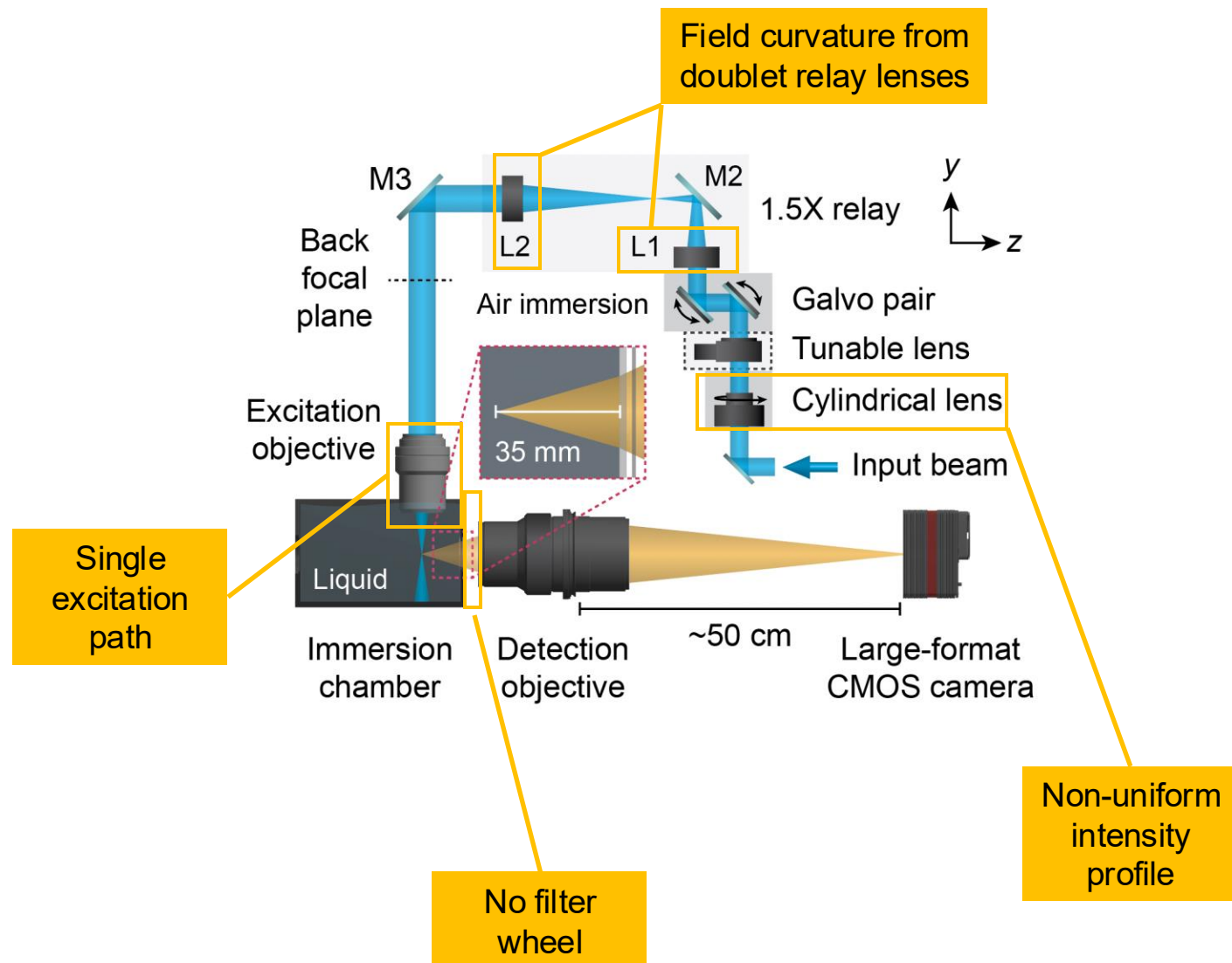
- 1 micrometer scale resolution
- 2 minimal tiling
- 3 large volumes
- 4 high isotropy
- 5 fast imaging speed
- 6 negligible aberrations

Leveraging electronics metrology technologies



Metrology technologies can be **powerful tools**
for **biological microscopy**

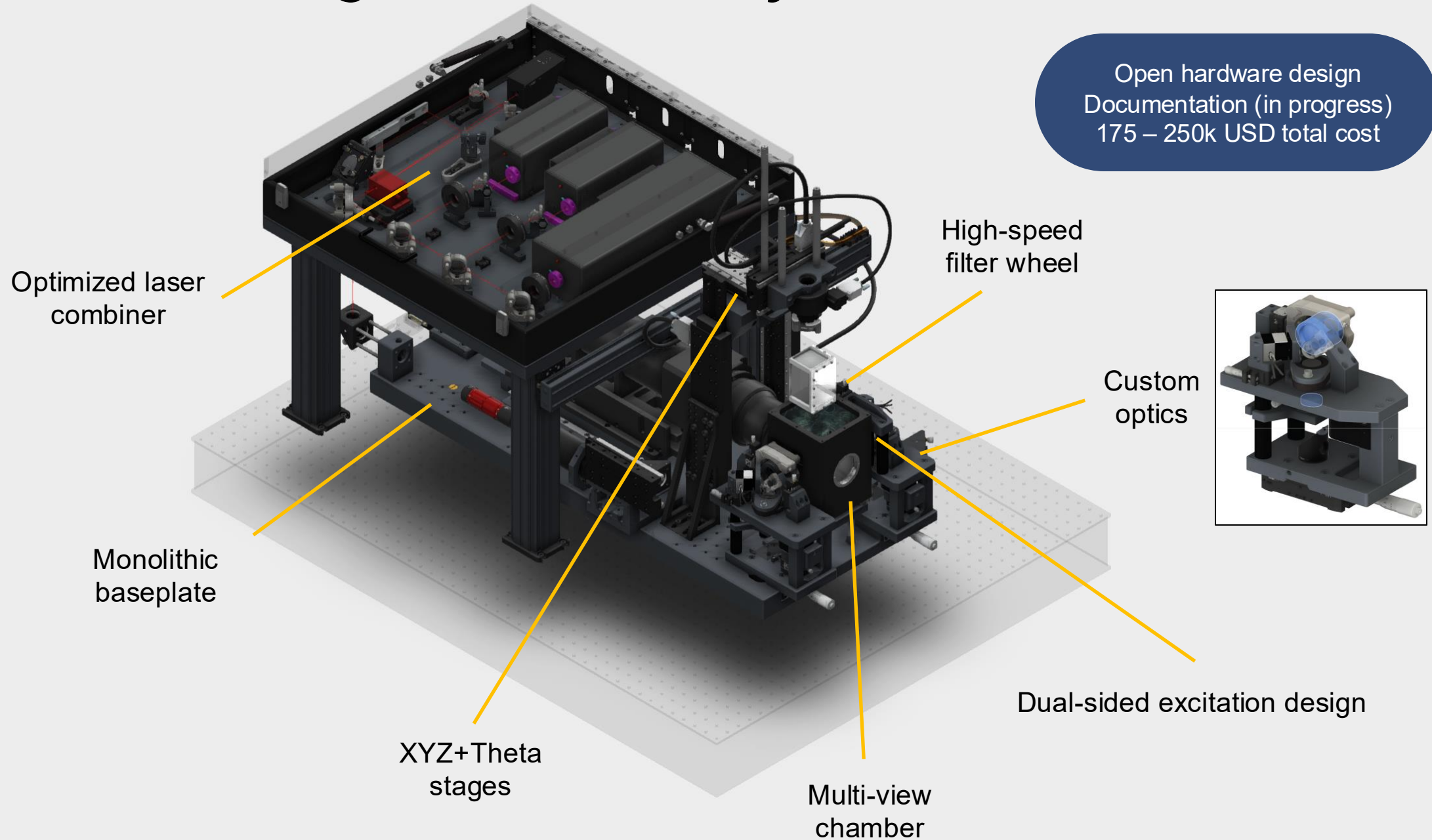
Prototype - ExA-SPIM microscopy system



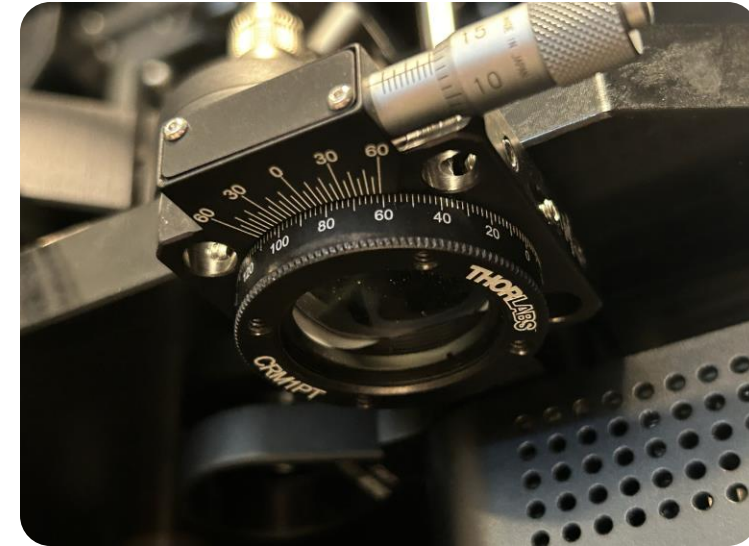
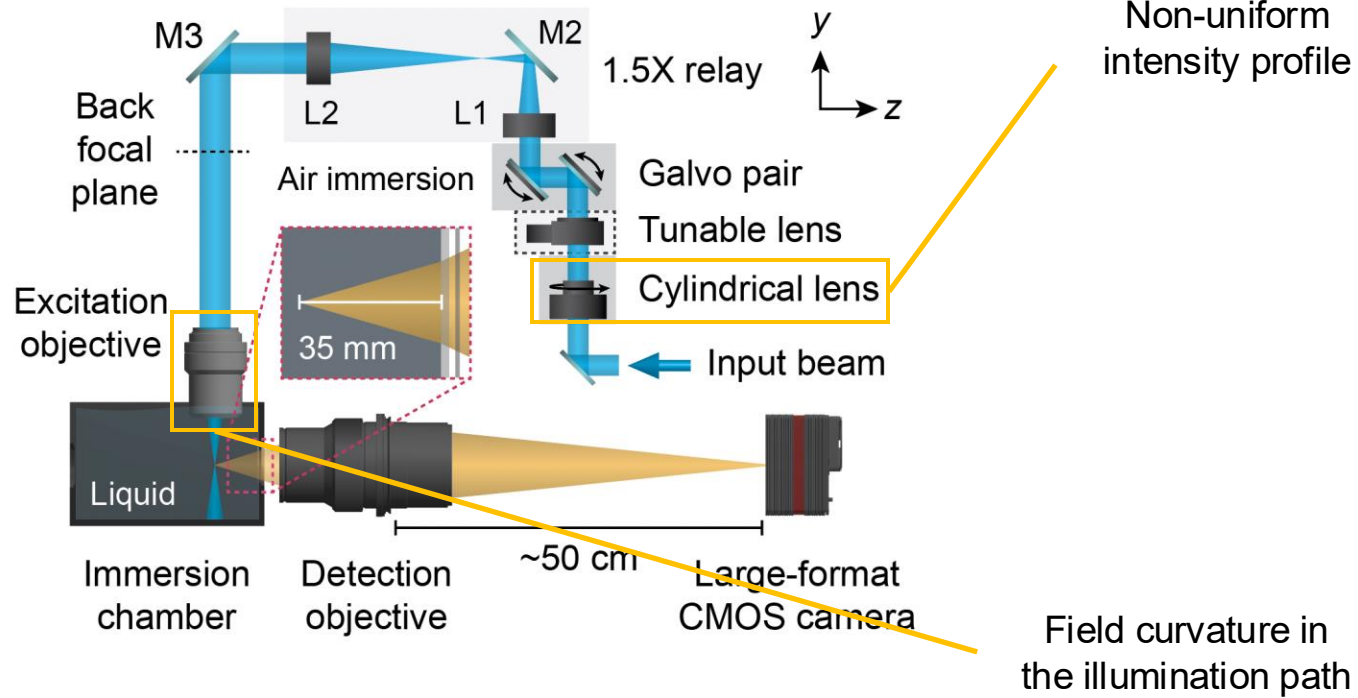
Thorlabs based system is cumbersome to build and align

Optimized re-designed ExA-SPIM system

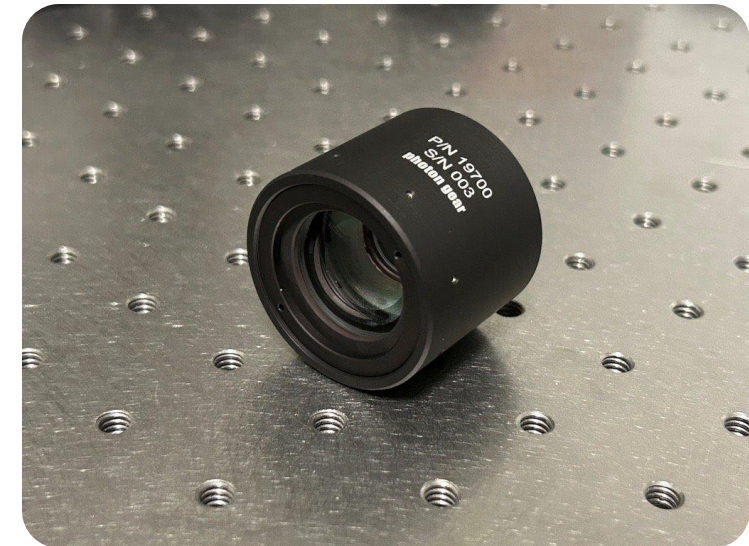
Open hardware design
Documentation (in progress)
175 – 250k USD total cost



Two main areas of improvement

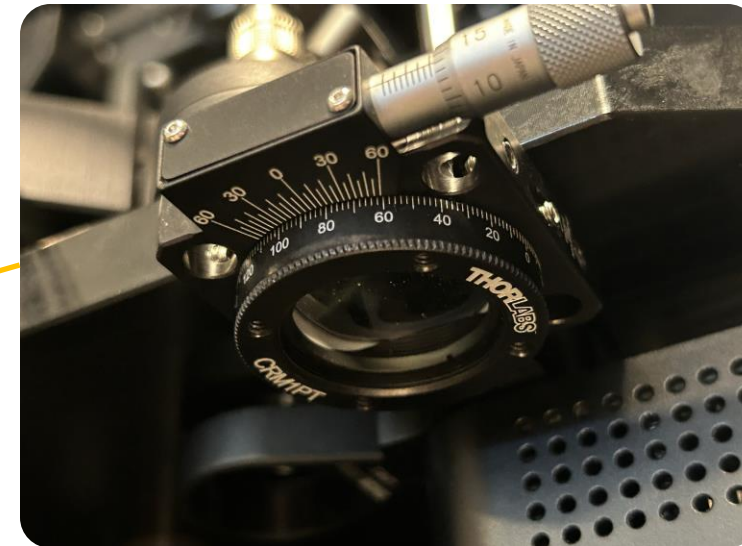
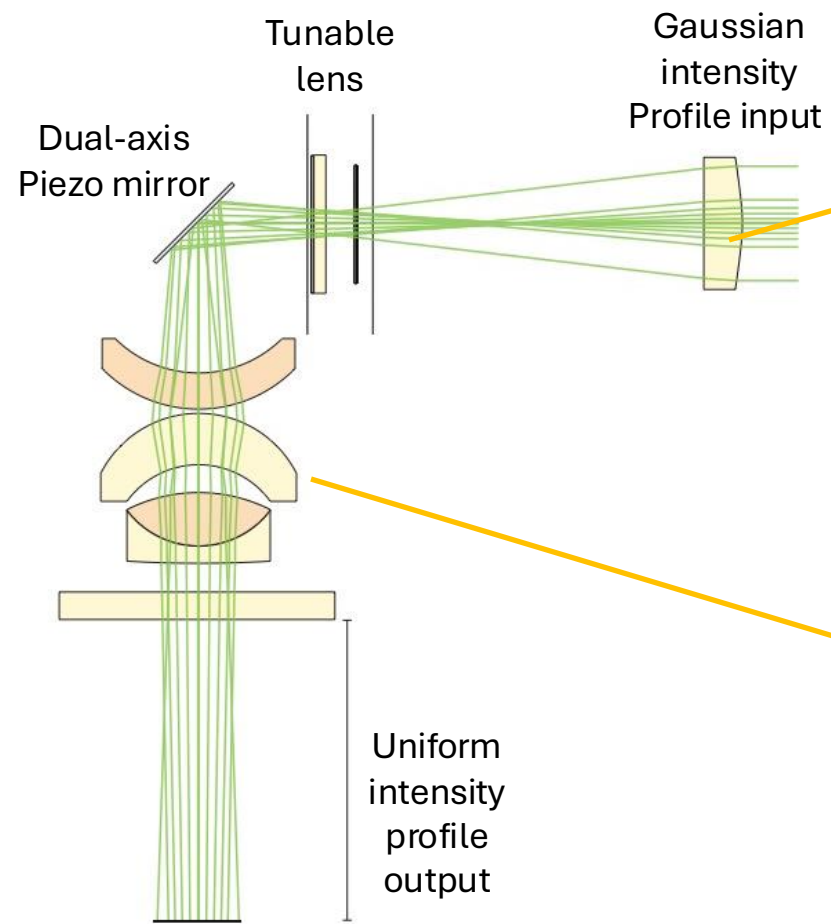


Custom
Powell lens
OPTIMAX[®]



Custom
objective
NA = 0.13
FOV = 12 mm
WD = 50 mm
photon gear

Two main areas of improvement

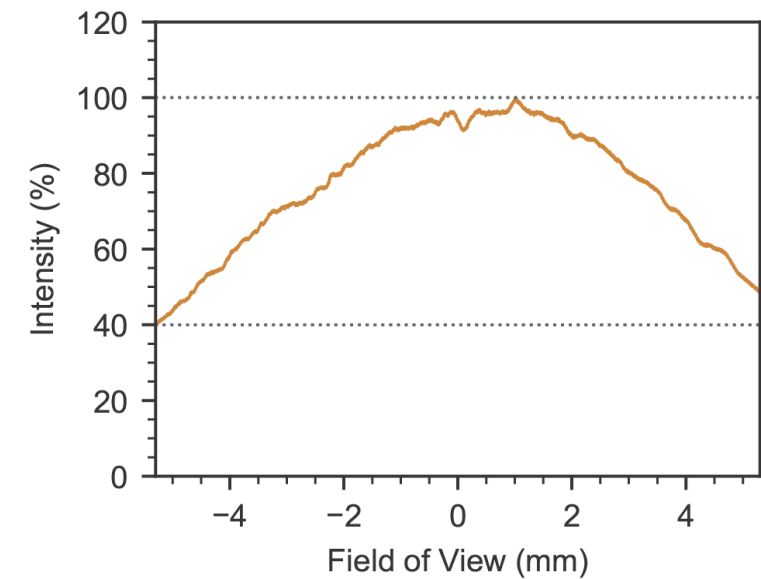
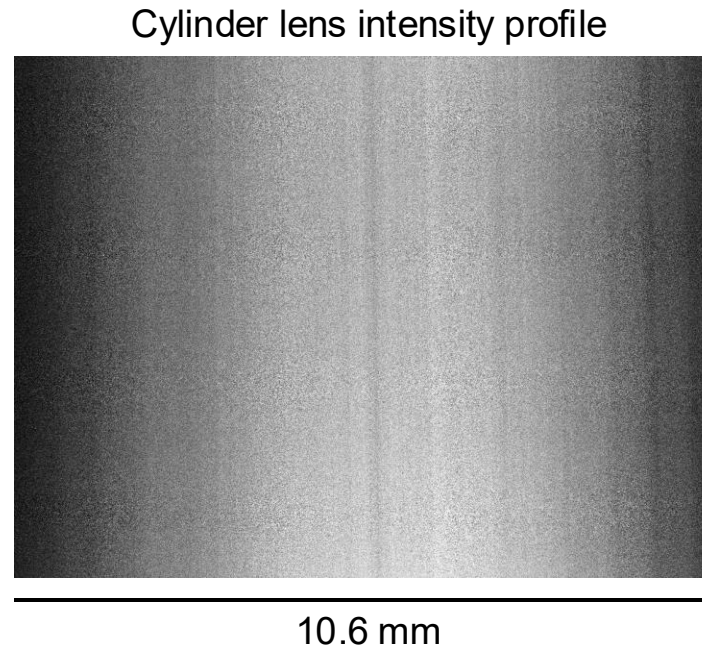
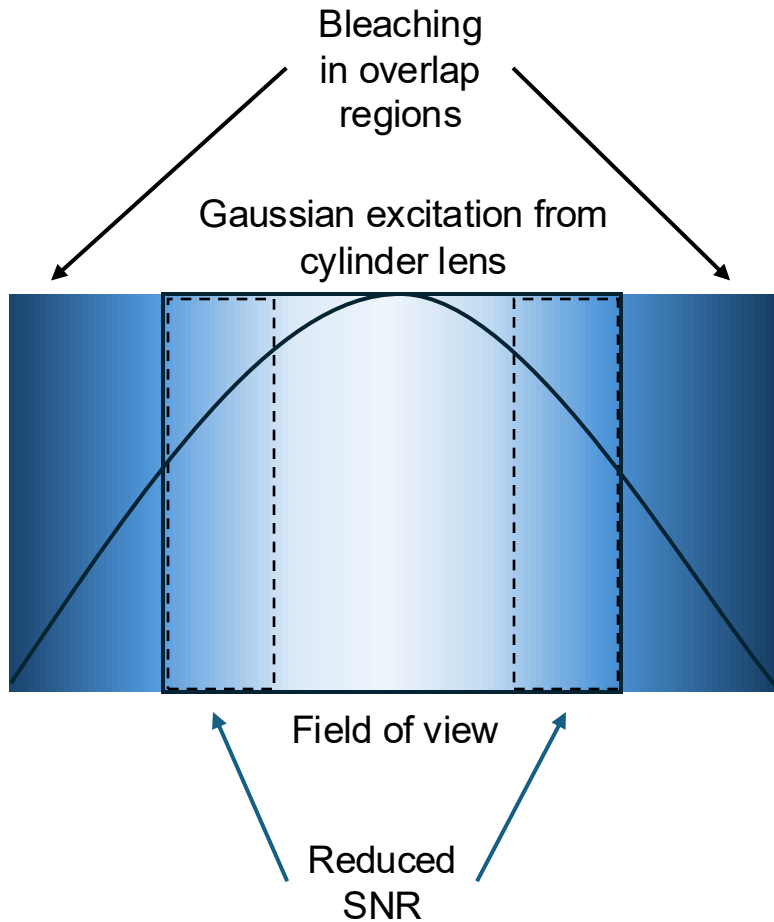


**Custom
Powell lens**
OPTIMAX[®]

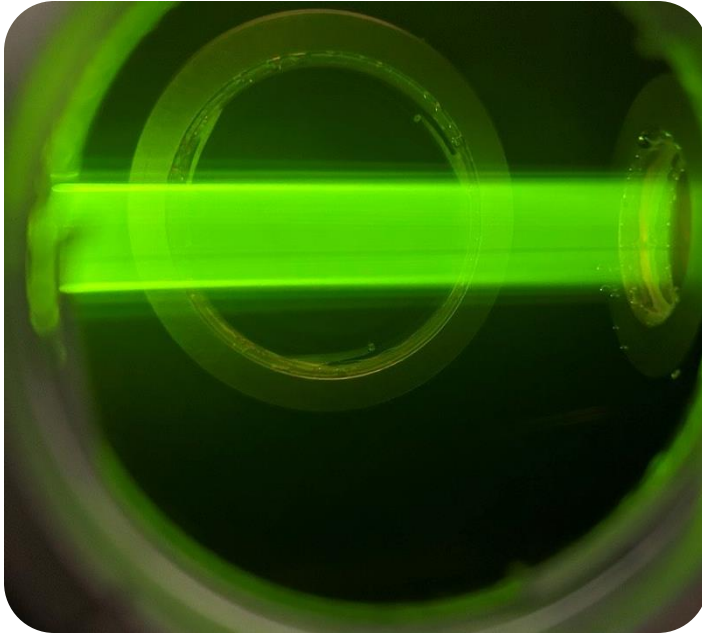


**Custom
objective**
NA = 0.13
FOV = 12 mm
WD = 50 mm
photon gear

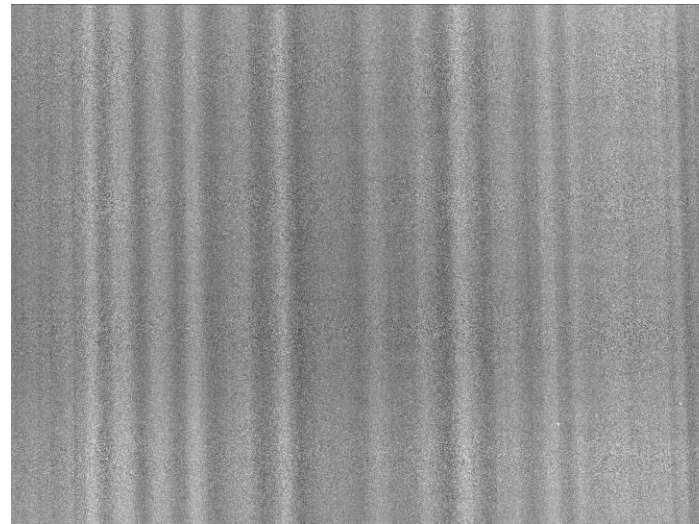
Non-uniform illumination



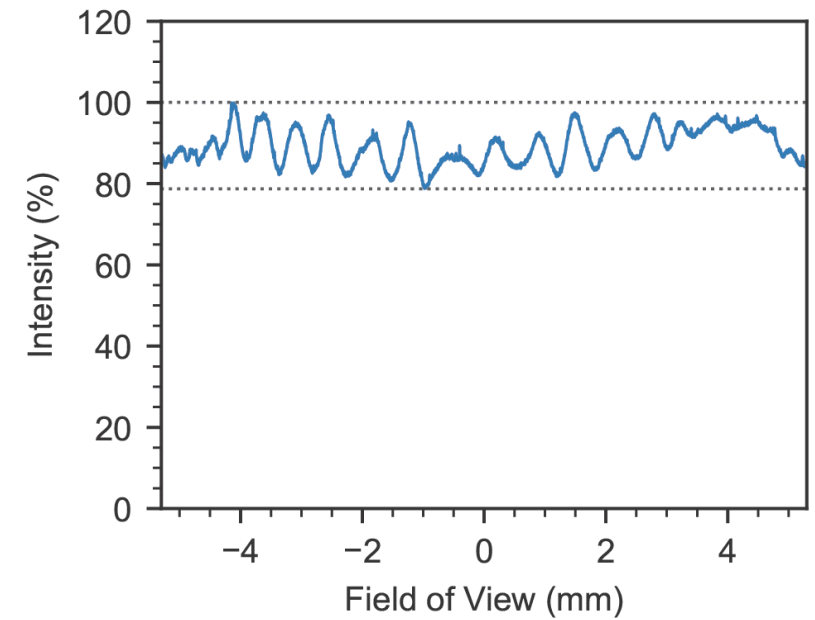
Powell lens characterization



Powell lens intensity profile

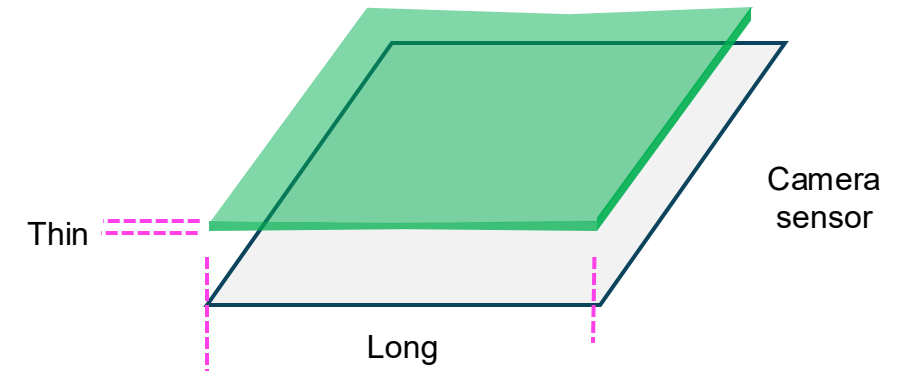
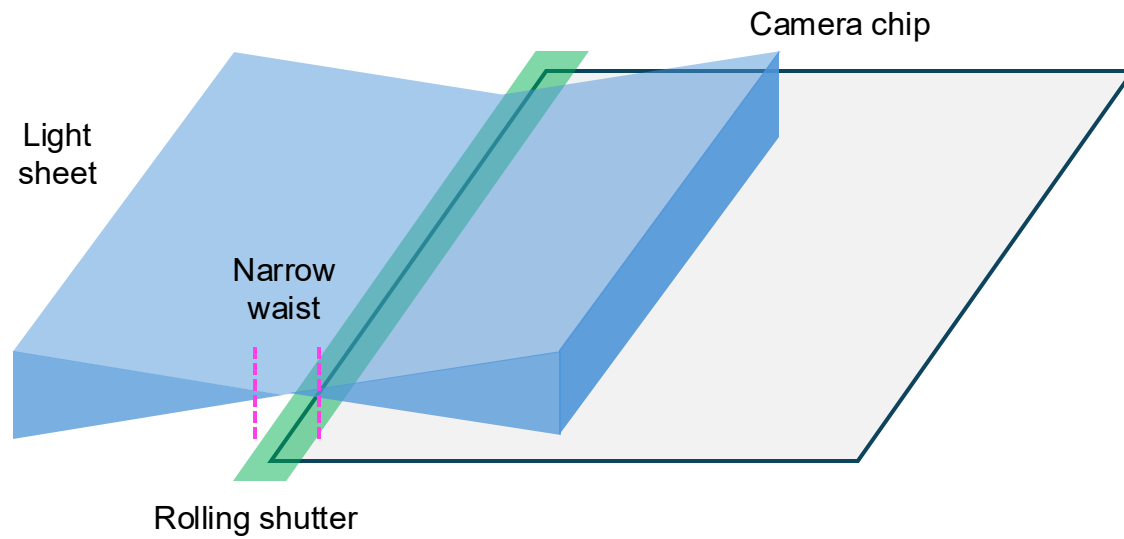
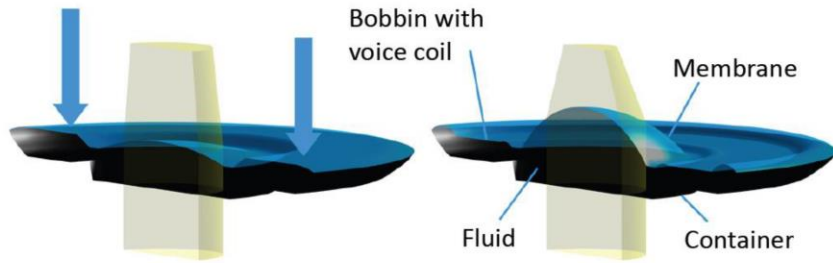


10.6 mm

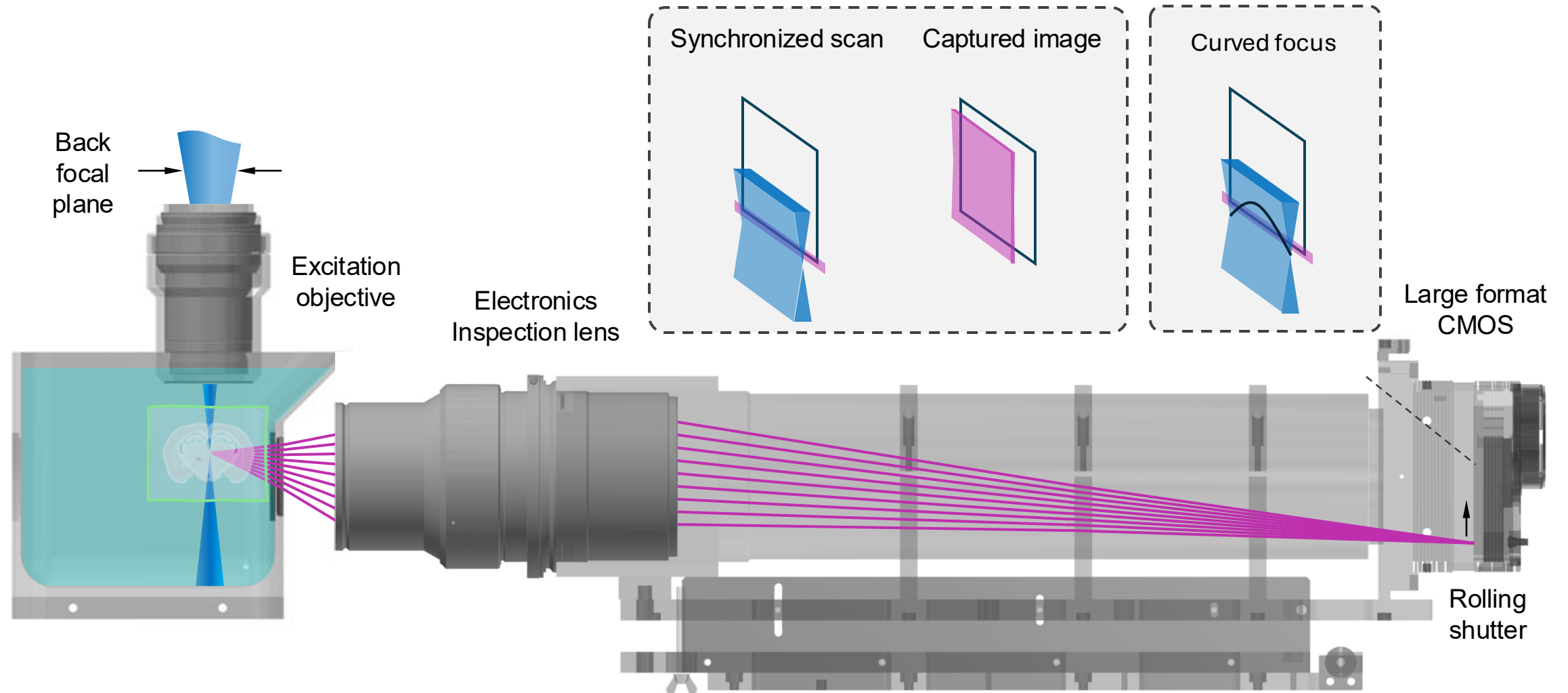


Axial sweeping

Electrical tunable lens



Axially-swept light-sheet microscopy (ASLM)

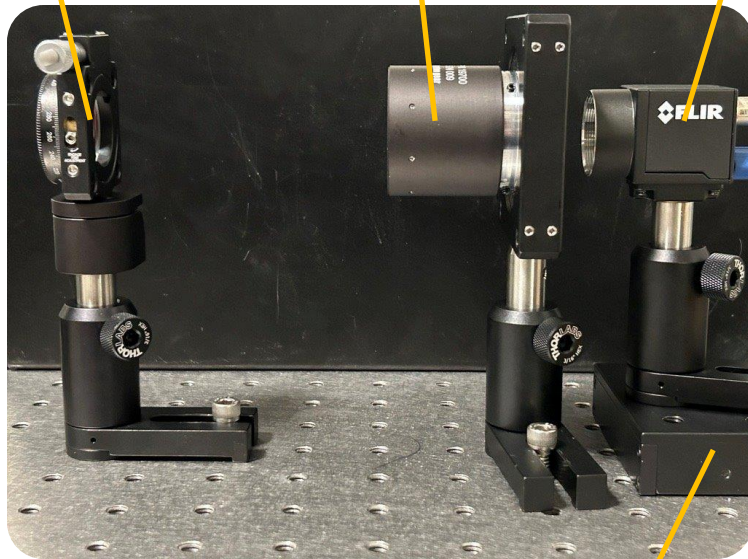


Objective lens characterization

Powell lens

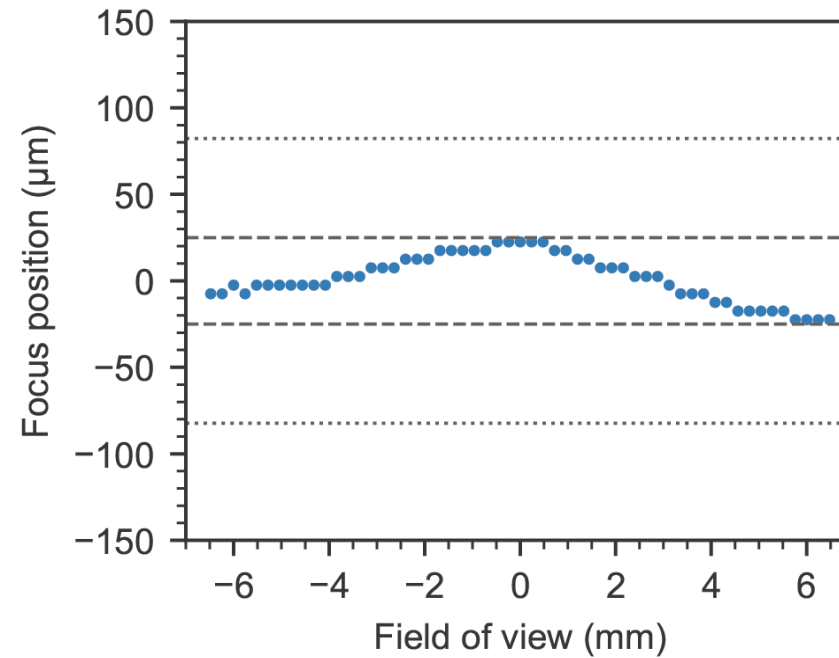
Custom Objective

Camera



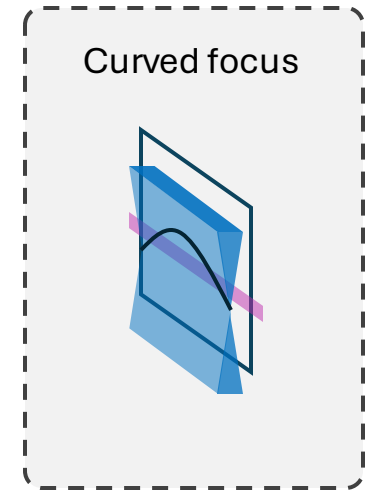
Motorized linear stage

Field curvature across FOV

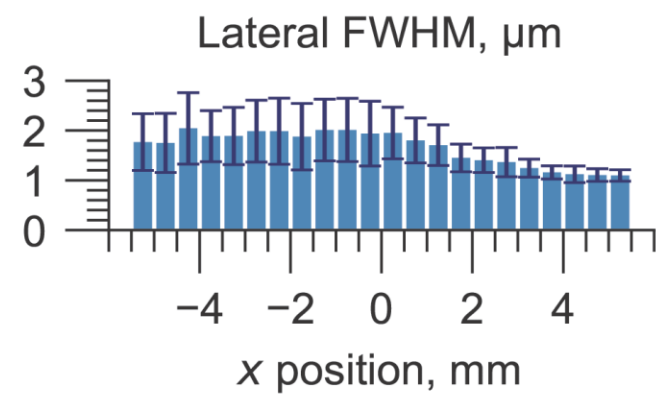
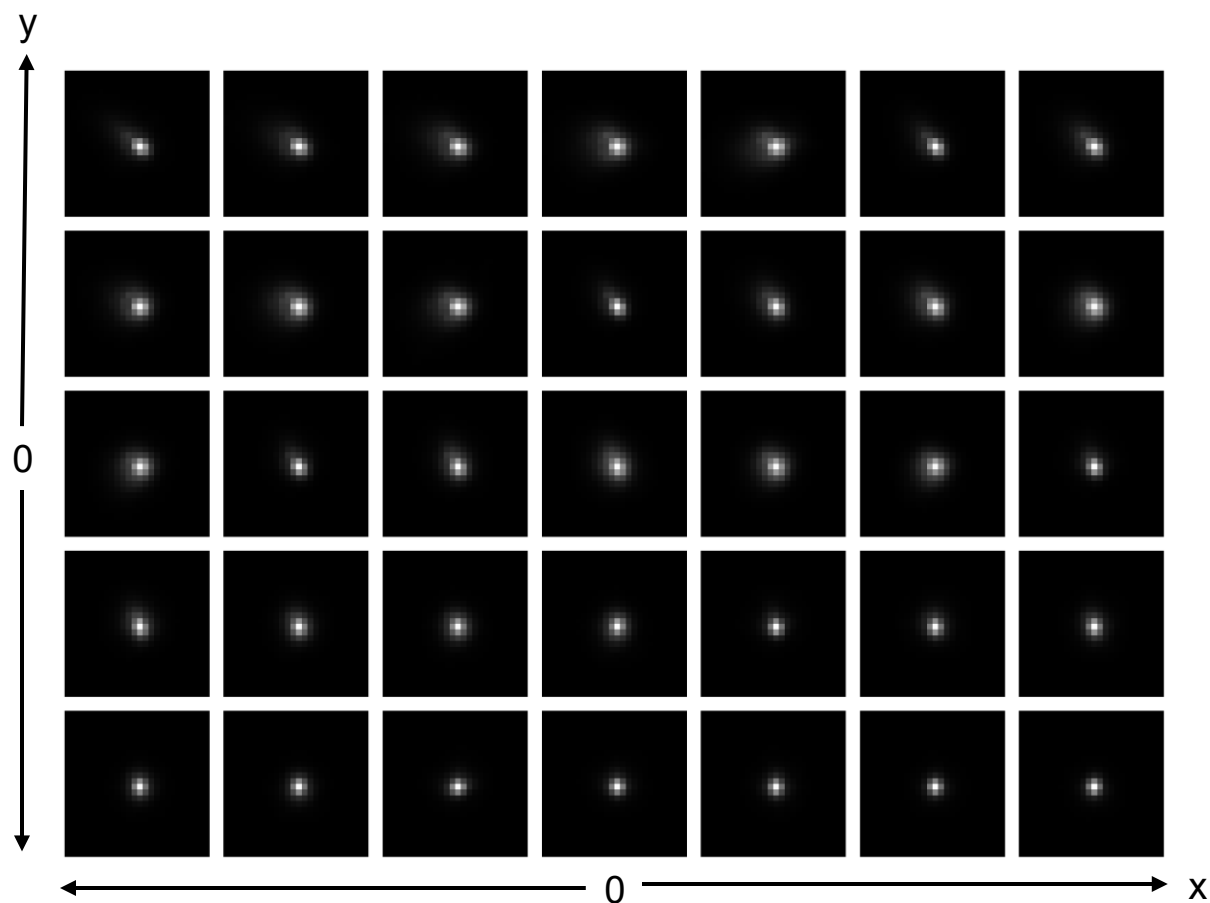


~50 μm field curvature

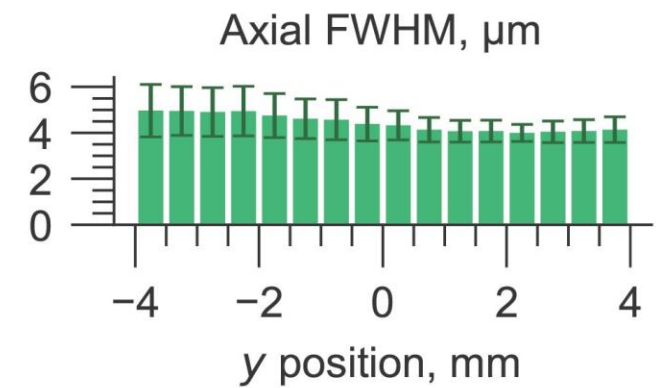
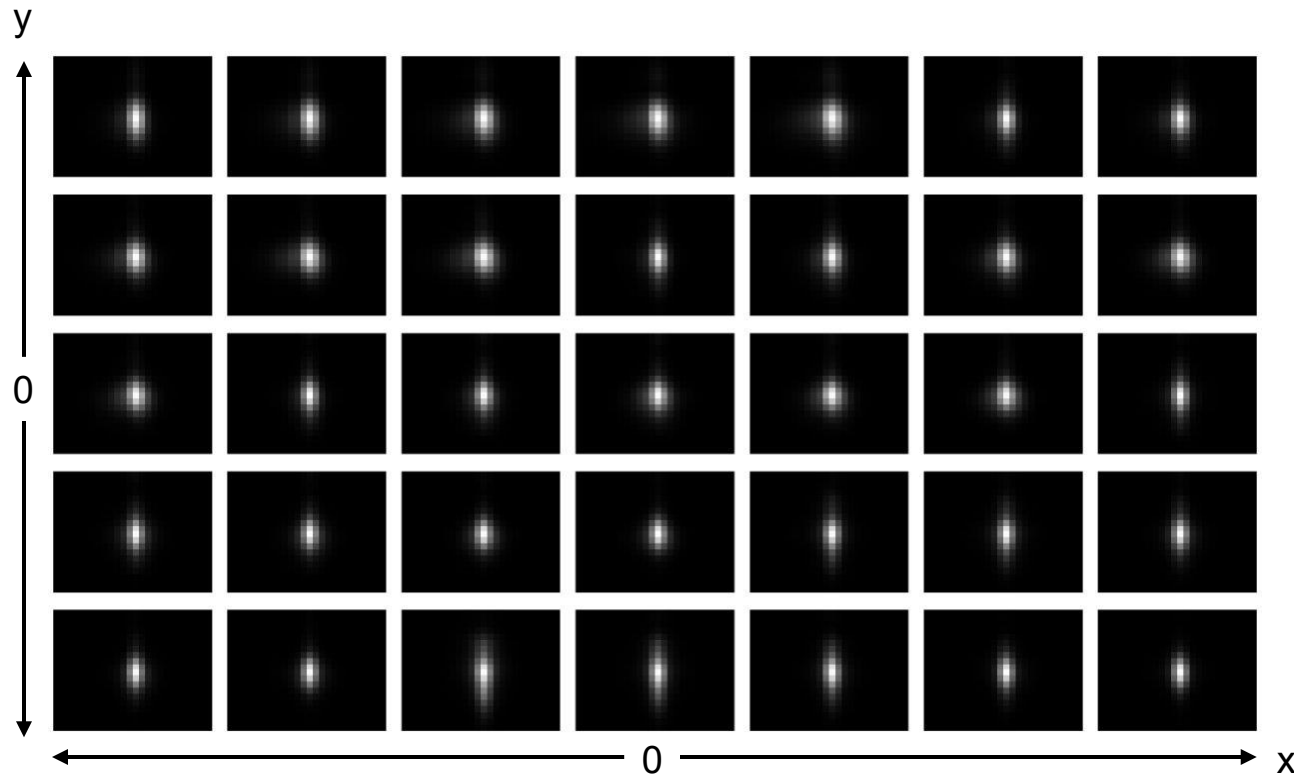
165 μm camera slit width



Lateral point spread function



Axial point spread function



Effective resolution: 0.25 x 0.25 x 0.40 μm

Expansion factor: 3X

Imaging time: 24 hr

Dataset size: 94 TB

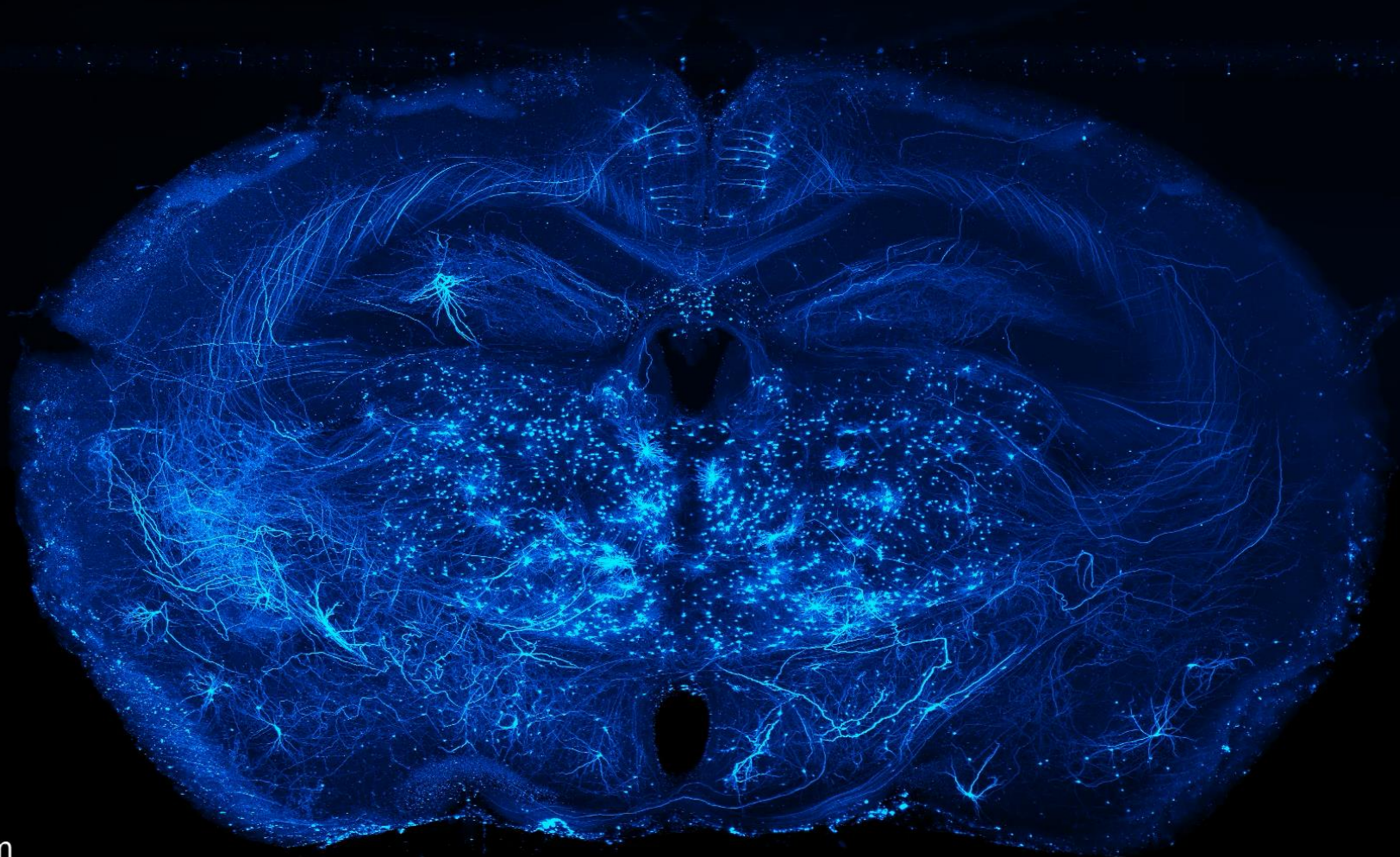
Wavelength(s): 488 nm

Mouse line: VGLuT2 (Slc17a6)



5 mm

zoom-in

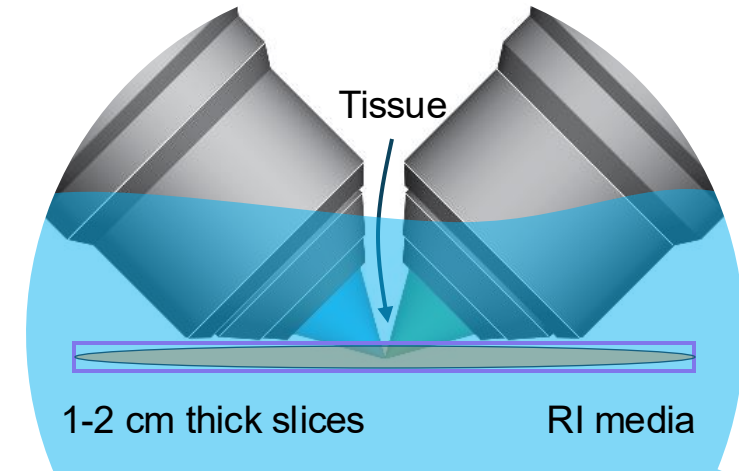
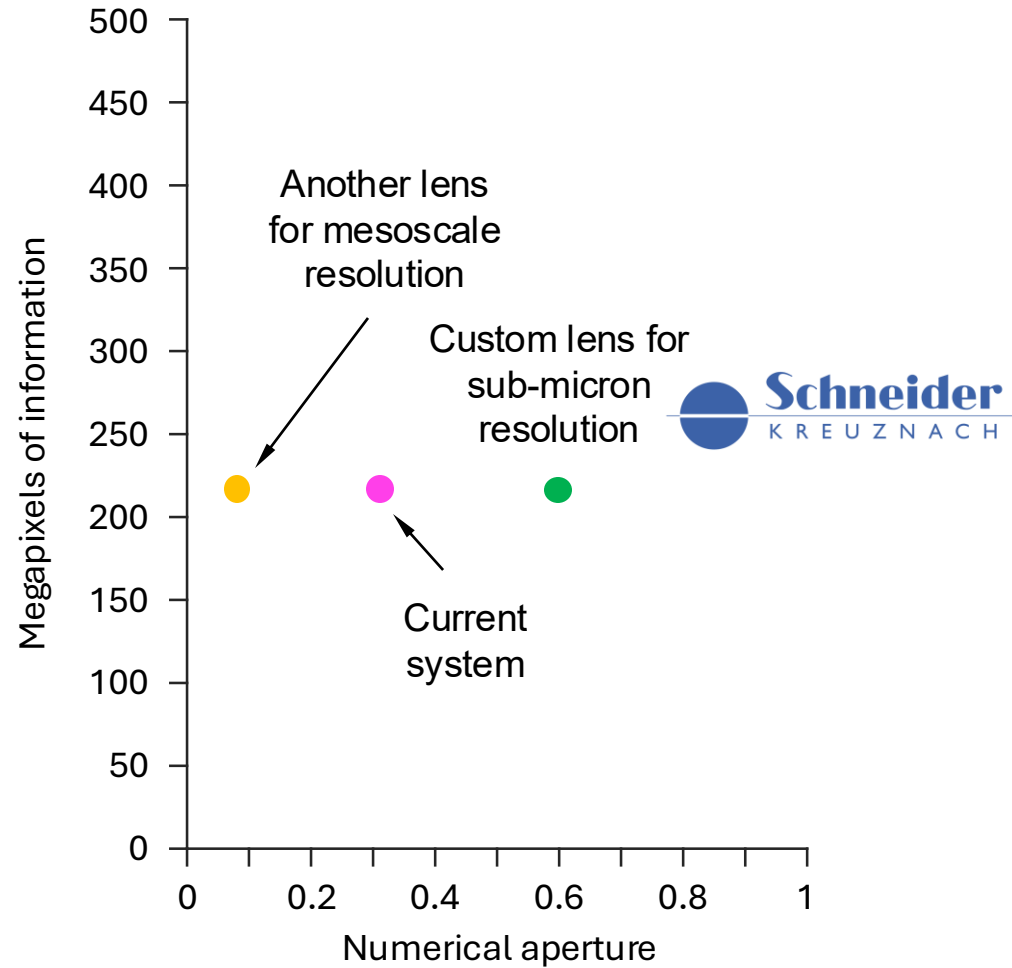


2 mm



A substrate for future innovation

Inverted SPIM for larger mammalian brains



System specifications	
Native resolution (μm)	
Field of view (mm)	
Working distance (mm)	
Throughput (cm^3 per day)	
Human brain (days)	

Open source documents will be available in **mid 2025**
Designing other system variations through **early 2026**
These tools can be applied to other scientific questions!

THANK YOU

Leadership

Karel Svoboda

Molecular Anatomy

Adam Glaser

Jayaram Chandrashekar

Cameron Arshadi

Micah Woodard

Walter Mwaniki

Xiaoyun Jiang

John Rohde

Sonya Vasquez

Naveen Ouellette

Judith Baka

Kevin Cao

Anna Grim

Polina Kosillo

Molly Logsdon

Mathew Summers

Holly Myers

Rachel Flannery

Laura Roy

Rajvi Javeri

Reconstructions

Peter Grotz

Phil Lesnar

Laila El-Hifnawi

Yun Wang

+ more

Scientific Computing

Saskia DeVries

David Feng

Sharmishta Seshamani

Gabor Kovacs

Jonathan Wong

Camilo Laiton

Brain Science

Mike Taormina

Jack Waters

Bosiljka Tasic

Shenqin Yao

Marcus Hooper

Administrative

Jim Berg

Lisa Kim

Gabi Martins

Shelby Suckow

Abbie Weaver

Tamara Zeric

CZI Science

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Alan Lidell

Janelia Research Campus

Dan Flickinger

Brian Coop

